E-Commerce Customer Intelligence System

Project Summary

💠 Overview

This project demonstrates an end-to-end **data analytics pipeline** designed to extract business insights from raw e-commerce data. It covers the complete workflow from data ingestion and cleaning to analysis, automation, and visualization.

💠 Objectives

- Transform raw transactional data into structured format
- Perform analytical queries to extract insights
- Automate report generation using Python
- Build an interactive dashboard for business decision-making

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💠 Tools & Technologies

- **Python** (Pandas, Matplotlib, SQLAlchemy)
- **SQL Server** (Joins, Aggregations, Window Functions)
- **Power BI** (Dashboard, DAX Measures, Data Modeling)
- **Jupyter Notebook** (EDA & Visualization)

💠 Workflow

1. **Data Cleaning (Python)**
    - Handled missing values, incorrect data types, and inconsistencies
    - Generated a clean dataset (cleaned_data.csv)
2. **Database Integration (SQL Server)**
    - Created normalized tables (Fact & Dimension)
    - Loaded data using Python automation
3. **SQL Analysis**
    - Developed multiple queries for business metrics
    - Examples:
        - Monthly revenue trend
        - Top customers and products
        - Customer retention analysis
        - Payment method distribution

4. **Automation (Python)**
    - Automated execution of SQL queries
    - Exported results to CSV and Excel reports
5. **Exploratory Data Analysis (EDA)**
    - Identified trends in revenue, customers, and products
    - Generated visual insights using Matplotlib
6. **Dashboard (Power BI)**
    - Built interactive dashboard with KPIs and filters
    - Enabled real-time business insights

◈ Key Insights

- Revenue shows consistent growth over time
- A small set of products drives the majority of revenue (Pareto effect)
- Business is highly dependent on high-value (VIP) customers
- Digital payments dominate customer transactions
- Sales peak during specific hours of the day
- Regional performance varies significantly

◈ Business Impact

- Helps identify high-performing products and customers
- Supports targeted marketing strategies
- Enables data-driven decision-making
- Improves customer retention planning

◈ Conclusion

This project showcases a complete **data analytics solution**, integrating Python, SQL, and Power BI to convert raw data into actionable business insights.

◈ Author

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